Noah Burnette

I make software for the manufacturing industry

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I am a software engineer who helps digitally transform manufacturing around the world.

EXPERIENCE

Industrial Software Engineer · Intellic Integration — Dec 2023 – Present

- Led development of large-scale, distributed industrial applications using Ignition and Unified Namespace (UNS) architecture.
- Applied domain-driven design (DDD) principles to model complex operational and business workflows.
- Designed and implemented core developer infrastructure, including a custom debugging library, lightweight ORM, and automated test discovery library.
- Developed Python microservices to support client-specific operational requirements.
- Built internal tooling in Go and Python to accelerate deployment workflows and improve engineering efficiency.
- Designed and implemented CI/CD pipelines to automate deployments and reduce regression risk.
- Mentored junior engineers through code reviews and architectural discussions, increasing code quality and engineering standards.
- Authored comprehensive technical documentation to support stakeholder adoption and long-term system maintainability.
- Delivered solutions across multiple industrial sectors, including food & beverage and battery manufacturing.

Software Developer Intern · Sierra Nevada Brewing Co. — Apr 2023 – Nov 2023

- Developed warehouse and quality-related systems using Ignition.
- Wrote and optimized SQL stored procedures and views for inventory and laboratory reporting.
- Implemented backend functionality in Python for data processing and enforcement.
- Developed operator-facing interfaces to streamline data entry processes.

Network Technician · Microtech Knives — Dec 2021 – Apr 2023

- Deployed and maintained Linux-based infrastructure supporting internal systems.
- Built and maintained Docker images to standardize development and workstation environments.
- Performed network traffic analysis to identify vulnerabilities and troubleshoot connectivity issues.
- Configured and managed Meraki networking hardware, including firewalls, switches, and access points.

PROJECTS

Ignition Apache Camel · Java module that runs Apache Camel routes inside Ignition 8.3. — Nov 2026 – Present

- Embedded Apache Camel in Ignition so developers can connect manufacturing data to external systems through YAML or Jython routes without deploying a separate integration service.
- Built adapters for Ignition tags, scripts and messaging, bringing platform-specific operations into Camel's routing framework.
- Designed reusable connection and certificate configuration with separate runtimes per integration, allowing shared settings without sharing live runtime objects.

slowrm · A lightweight ORM for Ignition. — Jul 2026 – Present

- Bridged SQLAlchemy's query compiler with Ignition's database APIs, enabling model-based persistence through existing Gateway database connections.
- Implemented object identity tracking, change detection and transaction-scoped sessions to replace repetitive persistence code and roll back uncommitted work automatically.

Ignition Package Maker · A CLI for packaging Ignition script modules as Python packages. — Jun 2026 – Present

- Built a Python CLI that converts Ignition project scripts into distributable packages, enabling shared libraries to be versioned and installed with pip.
- Preserved module import paths and package metadata across rebuilds, allowing developers to maintain scripts in Ignition while distributing them through standard Python tooling.

Ignition Sift (VS Code Extension) · Barebones, stub-driven auto-imports for Ignition scripting projects. — Dec 2025 – Present

- Built a VS Code extension that generates Python type stubs from Ignition scripts, making project APIs available to Pyright without changing the deployed code.
- Translated Python 2 type comments into annotated signatures and indexed symbols for Quick Fix imports, reducing manual lookup of nested script paths.

SKILLS

Languages	Python, HTML/CSS, Java, TypeScript, SQL, Go
Frameworks/Libraries	FastAPI, Apache Camel, Ignition SDK, JUnit, Pytest, React, htmx, Node.js
Platforms/Tools	Git, Ignition, Docker, Kubernetes, MSSQL, PostgreSQL
Architectures/Practices	Unified Namespace, Domain-Driven Design, Test-Driven Development, Event-Driven Architecture, Microservices Architecture, CI/CD

EDUCATION

University of North Carolina Asheville · Bachelor of Science, Computer Science, GPA 3.78 — May 2020 – Dec 2023